Input Ontology (IO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *CRISISLTLSUM* | ***Context:*** *Ecuador — Humanitarian Crisis Social Media Analysts*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark restricts itself to four crisis domains — wildfire, local fire, traffic, and storm [Q122] — and explicitly defines events as 'local' incidents 'bound to an exact location, such as a building, a street, or a county, and usually lasts for a short period' [Q12]. The Ecuador deployment's priority crisis types are volcanic eruptions, earthquakes, landslides, coastal/riverine flooding, mass displacement, and complex humanitarian emergencies — none of which appear in the benchmark taxonomy. The elicitation explicitly identifies large-scale humanitarian crises that 'exceed local response capacity' as the operational priority, structurally incompatible with the benchmark's local/short-duration framing. Storms have only partial overlap, and even there the benchmark's storm category was selected for North American contexts (Texas/Florida hurricanes, NSW storms) [Q124], not Andean lahares or tropical Pacific precipitation. An independent Ecuadorian Twitter monitoring system identified four distinct categories — volcanic, telluric (seismic), fires, and climatological — confirming that Ecuador's operational taxonomy diverges fundamentally [WEB-18].

ID	Evidence
Q12	“Moreover, we focus on the extraction of local crisis events. The term "local" indicates that an event is bound to an exact location, such as a building, a street, or a county, and usually lasts for a short period.” (p.2)
Q122	“We build CrisisLTLSum by limiting the search to four crisis domains: wildfire, local fire, traffic, and storm.” (p.12)
Q124	“These states are selected as they are some of the areas subject to the most events falling into our crisis domains of interest. California, Victoria, and New South Wales are mainly selected due to the abundance of wildfires in hot seasons. Texas and Florida have been selected as they are both subject to wildfires and storm events during different months. Massachusetts and Illinois are selected first because those areas are frequently subjected to bad weather, and second as they contain big cities subject to traffic and local fire events alongside New York.” (p.13)
WEB-18	An Ecuador-specific Twitter monitoring system independently identified four distinct crisis categories — volcanic, telluric (seismic), fires, and climatological — diverging from the benchmark taxonomy (https://www.researchgate.net/publication/312485537_System_for_monitoring_natural_disasters_using_natural_language_processing_in_the_social_network_Twitter)

Strengths

- The taxonomy does include 'storm' and 'local fire' categories that have partial conceptual overlap with páramo fires and some Costa flooding scenarios in Ecuador [Q122, Q124]; the 'local' event framing [Q12] could apply to canton-level fire events in urban Ecuadorian contexts.

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Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for Ecuador humanitarian deployment include volcanic eruptions and lahares (Cotopaxi, Tungurahua, Sangay), earthquake/aftershock sequences (Manabí 2016 type), landslides (movimiento en masa), coastal/riverine flooding (desbordamiento), mass displacement (including Venezuelan migration and volcanic exclusion zones), and complex humanitarian emergencies. The elicitation confirms these prioritized categories and notes they fall outside the benchmark's four-category set.

IO-2: Yes — the benchmark omits all of Ecuador's priority geophysical and humanitarian crisis types. The authors confirm coverage is limited to wildfire, local fire, traffic, and storm [Q122], with no volcanic, seismic, landslide, displacement, or complex-emergency content.

IO-3: Traffic incidents and US-style wildfire and storm categories are not operationally irrelevant in Ecuador per se but are tuned to a US/Australian event profile (Texas/Florida storms, California/NSW/Victoria wildfires [Q124]) that does not match Ecuadorian risk geography.

IO-4: The benchmark structurally cannot evaluate model performance on volcanic eruption timelines, earthquake aftershock sequences, displacement events, or humanitarian emergencies. This is a content-validity violation for the deployment: the construct of 'crisis timeline' the benchmark measures is a strict subset of, and partially disjoint from, the construct relevant to Ecuadorian humanitarian coordinators.

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WEB-21	HumVI review notes humanitarian event detection datasets disproportionately English; structural relevance categories for humanitarian work are underrepresented (https://arxiv.org/html/2410.06370)

Information Gaps

- Whether storm-category models trained on US/Australian data have any zero-shot transfer to Ecuadorian flooding or páramo storm contexts is empirically unknown.

Requires Expert Verification

- The exact distribution of crisis types in the Ecuadorian humanitarian tweet stream (proportion volcanic vs. flooding vs. displacement) requires field stakeholder input from Cruz Roja Ecuatoriana / OCHA Ecuador.

Remediation

Gap 1: Benchmark covers only wildfire, local fire, traffic, and storm; deployment priorities (volcanic, seismic, landslide, flooding, displacement, complex emergencies) are entirely absent.

Recommendation: Define an Ecuador-specific crisis taxonomy aligned with SNGR alert categories and IGEPN volcanic phases; do not use CrisisLTLSum scores as evidence of model capability on Ecuadorian disaster types.